Genet Resour Crop Evol (2018) 65:449–469
https://doi.org/10.1007/s10722-017-0544-y

RESEARCH ARTICLE

Trends and drivers of on-farm conservation of the root legume ahipa (*Pachyrhizus ahipa*) in Bolivia over the period 1994/96–2012

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Received: 2 January 2017 / Accepted: 11 July 2017 / Published online: 27 July 2017
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Abstract The leguminous Andean root crop ahipa has become scarce and the current status of in situ conservation in Bolivia is concerning. Agro-biodiversity changes affect the use, conservation and socio-economic trends for ahipa (*Pachyrhizus ahipa*) and to substantiate this, a comparison of two cropping periods were made. Aspects of root production and the shift from cultivation of ahipa to cash crops such as maize were compared for two periods: (a) 1994/96 and (b) 2012. Our study showed that the price of ahipa had not increased in 70% of the urban markets; hence, there is little incentive to retain the cultivation of ahipa. We found that rural ahipa growers selected the largest seeds, but did not select seeds by colour. The mixed seed colour predominated the conservation of ahipa. Laborious yield enhancing practices required in field management and low market value endanger future conservation of the ahipa. We conclude that additional research is needed in order to safeguard the extant variation of ahipa, i.e. nutritional value, taste, stress tolerance and market potential.

Keywords Andean yam bean · *Pachyrhizus ahipa* · Orphan food crop · Endemic leguminous root crop · Inter-Andean valleys · Organic farming

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Introduction

The conservation of *on-farm* agro-biodiversity—especially involving local landraces and underutilized crops—is threatened by monoculture and marketable crops as farmers are confronted by income decline due to reduced demands for their products (Galluzzi and López-Noriega 2014a, b; Jacobsen et al. 2013, 2015; Khoury et al. 2014; INE 2015). The conservation of on-farm agro-biodiversity is commonly associated with remote rural regions (Chivenge et al. 2015; Nyadanu et al. 2016). Thus the consumer attention is often limited (Jarvis et al. 2000). In Latin America, the numerous native Andean root and tuber crops—domesticated one to two millenia before present—are good examples of this situation (National Research Council, 1989; Flores et al. 2003; Pickersgill

2007; Galluzzi and López-Noriega 2014a). For example, the Andean smallholder farmers still cultivate ahipa (*Pachyrhizus ahipa* (Wedd.) Parodi) (Cadima et al. 2009; Lopez et al. 2012), consumed almost exclusively by rural consumers and by migrants in a few urban communities (Ørting et al. 1996; Bjerre 2000; Flores 2001; Gruca 2010; Hurrell et al. 2011; Contreras 2012; Pochettino et al. 2012; Hurrell and Pochettino 2014; Puentes and Hurrell 2015). The product, the edible tuberous root, is usually scarce in Bolivian urban markets and is thus at risk of disappearing completely. White maize cobs (USD 0.36 kg^{-1}) and tomato (USD 0.35 kg^{-1}) are more marketable selling at twice to three times the price of ahipa (USD 0.63 kg^{-1}) and can provide income to farmers (Candela 2013; Rivera 2014). Given this situation, ahipa must be considered a marginal crop (Galluzzi and López-Noriega 2014a, b). The edible root is—due to its fruity, crunchy and juicy quality—regarded as a seasonal 'fruit'rather than a 'proper' root crop (Doporto et al. 2014). The crop is seed propagated, and seeds are often stored under inadequate conditions (Ørting 1996; Gruca 2010). Ahipa seeds are exchanged via informal seed systems within rural communities (Bjerre 2000). When farmers exchange seeds informally, root characteristics are used to describe the plants (Flores 2001; Gruca 2010). Usually, roots are commercialized for the urban Catholic celebration of Corpus Cristi as this constitutes a good market opportunity (USD 0.44–0.73 kg^{-1}) (Contreras 2012; Leon 2015).

The Andean yam bean was first described by Weddell following a visit to an urban market in La Paz (Weddell 1857). Furthermore, in 1883 Campos identified the purple mottled fleshy ahipa root in a remote area during an expedition from La Paz to Asuncion, Paraguay (Campos 1888). Parodi in 1935 mentioned its cultivation in a region located in northern Argentina and southern Bolivia (Parodi 1935, 1936). In 1966, Cárdenas (1989) reported ahipa cultivation in the Bolivian Inter-Andean-Valleys (IAVs). Later, in the 1980s Rea surveyed ahipa growers in the IAVs and the Yungas of La Paz (Rea 2004). At the 'Herbario Nacional Bolivia, Cota Cota' in La Paz, Bolivia, a few specimens of ahipa collected in Bolivia exist (Vidaurre et al. 2006), but prior to the first field study by Ørting (1994) only a few systematic investigations involving collection and conservation of ahipa in Bolivia had been conducted (pers. obs.).

In the early 1990ties, scientists collected samples of different yam bean species in Latin American countries (Sørensen 1988, 1996); however, the Andean yam bean, ahipa, was not included in this effort until Ørting's Bolivian field surveys in 1994 and 1996 (Ørting 1994, 1996). These surveys provided a preliminary national baseline of the crop and confirmed the previously available information (Cárdenas 1989; Ørting et al. 1996; Sørensen et al. 1997). Additionally, studies were conducted involving field and urban markets surveys in Chuquisaca and Tarija departments (Ørting 1994, 1996; Flores and Donaire 1999). No national initiatives to secure on-farm conservation of ahipa landraces have been implemented following Ørting's 1994 and 1996 surveys (Gruca 2010; Piispanen 2011; Galluzzi and López-Noriega 2014a, b). Albeit, according to Galluzzi et al. (2010) a collection of eighteen ahipa seed accessions is held by INIAF (Instituto Nacional de Innovación Agricola y Forestal) as a native crop, i.e. the only recent conservation initiative.

FAO recommended the monitoring and development of an early warning system to avoid loss of landraces for food and agriculture (FAO 2010, 2014). The on-farm monitoring of genetic erosion and landrace loss of ahipa in Bolivia is necessary because traditional cropping systems are rapidly disappearing together with the associated ethno-ecological knowledge. Socio-economic factors caused considerable modifications in the on-farm agro-biodiversity, and the quantification of the diversity at the in situ level can indicate the conservation status of valuable landraces (Jarvis and Hodgkin 2000; Jarvis et al. 2000, 2008a, b, 2011). The maintenance of genetic diversity within local production systems simultaneously helps to conserve local knowledge (FAO 2010). Several local surveys have been conducted in Bolivia over the last 20 years that indicated the decline in the number of both producers and local landraces (Bjerre 2000; Flores 2001; Ponce-Tovar 2002; Gruca 2010). It, therefore, became urgent to investigate nationally whether farmers continued to conserve the in situ genetic diversity of ahipa or if landraces have been irretrievably lost.

Two research questions were considered: (1) to what extent does the use of ethno-ecological knowledge and change in agro-biodiversity alter on-farm ahipa conservation; and (2) to what extent did the traditional agronomic practices and socio-economic

factors in the IAV rural communities of Bolivia affect the on-farm conservation of ahipa landraces in 1994, 1996 and 2012, respectively? We present and compare results from the two periods, and discuss the effect of changes in agro-biodiversity on ahipa on-farm conservation in eleven rural communities located in the IAVs.

Materials and methods

Study site

Ahipa landraces are cultivated in the IAV agro-ecological zone of the Bolivian departments of La Paz, Cochabamba, and Chuquisaca (Ørting 1994, 1996). This ecological zone features a prominent element of subsistence agriculture involving the use and conservation of Andean crops (Sims et al. 1999; Avila et al. 2012). Many rural communities are isolated far from urban centres; thus arrival at the sites frequently required overcoming considerable natural obstacles. One such community with ahipa cultivation is Lloja (Fig. 1A) where smallholder farmers grow vegetables and seasonal fruits. Other similar communities are Angostura and Ornuni (Fig. 1A) reported here for the first time as sites where ahipa in 2012 was found cultivated in pre-Columbian terraced fields on the banks of the La Paz River. In addition to the ahipa, farmers plant pulses, vegetables and other Andean tuber crops. The ahipa roots are commercialized only scantily in the urban markets of La Paz, and El Alto supplied from the communities of Anquioma Alto, Anquioma Bajo, and Asambo (Fig. 1A) located along the Porvenir and Luribay rivers in the Loayza Province (Ørting 1994). Here farmers harvest seasonal fruits, vegetables and cut flowers (Gruca 2010; Contreras 2012; Candela 2013). Regarding ethnic affiliation, all the above mentioned rural communities are Aymara-speaking.

Shifting to the Ayopaya River basin the communities here are located far from the rural town of Independencia in the Department of Cochabamba (Fig. 1A). In general, within the communities, the ahipa plots are located at some distance from the households. The ethnic Quechua-speaking farmers of this region also manage fruits, vegetable and Andean root and tubers. In Tarija city—in the southern department bearing the same name—the ahipa roots are very popular—more so than in other Bolivian cities (Ørting 1996; Flores and Donaire 1999; Gruca 2010). The ahipa roots marketed in Tarija originate from the Estancia Caraparí and the community of Pilaya near the Pilaya River in the department of Chuquisaca (Fig. 1B).

Data collection

The total area planted with each ahipa landrace was calculated based on GPS measurements and descriptions of their plots using the methods according to Jarvis and Campilan (2006). The distance between plants was recorded once or twice while the crop was in the flowering stage, i.e. depending upon the accessibility of the community and the farmer. The information collected on duration of the ahipa growth cycle and plant husbandry also included data regarding tasks after sowing and following flowering, e.g. reproductive pruning and root harvest. We further collected information on the number of other crop species managed by each farmer, as well as the overall crop diversity in each community. Changes in crop diversity among the ahipa landraces collected in each community were evaluated. We also recorded information on seeds of ahipa landraces from the farmers in 2012. The colour chart pattern of the Horticultural Royal Society (RHSL 2005) was used to classify seed testa colours. Besides, vernacular denominations of seed in the Aymara and Quechua dialects were registered according to information from farmers.

Interviews and agronomic surveys

The two field study reports by Ørting (1994, 1996) were used as data references to plan the 2012 field collections and to identify and to re-contact farmers—a.k.a. the *'ajiperos'*—a total of 105 farmers participated, i.e. 52 in 1994/96 and 53 in 2012 (Table 1). Farmers who grew or still grow ahipa landraces were contacted via the mediation of technicians (Table 1); however, due to logistics, no sampling procedure was carried out. The interviews were conducted in the IAV rural communities of the La Paz, Cochabamba, and Chuquisaca departments. Three farmers in the La Paz River basin community of Lloja were recorded by Ørting in 1994, but in 2012 only two were contacted, and additionally, four farmers participated in neighbouring Angostura and Ornuni in 2012. Farmers who

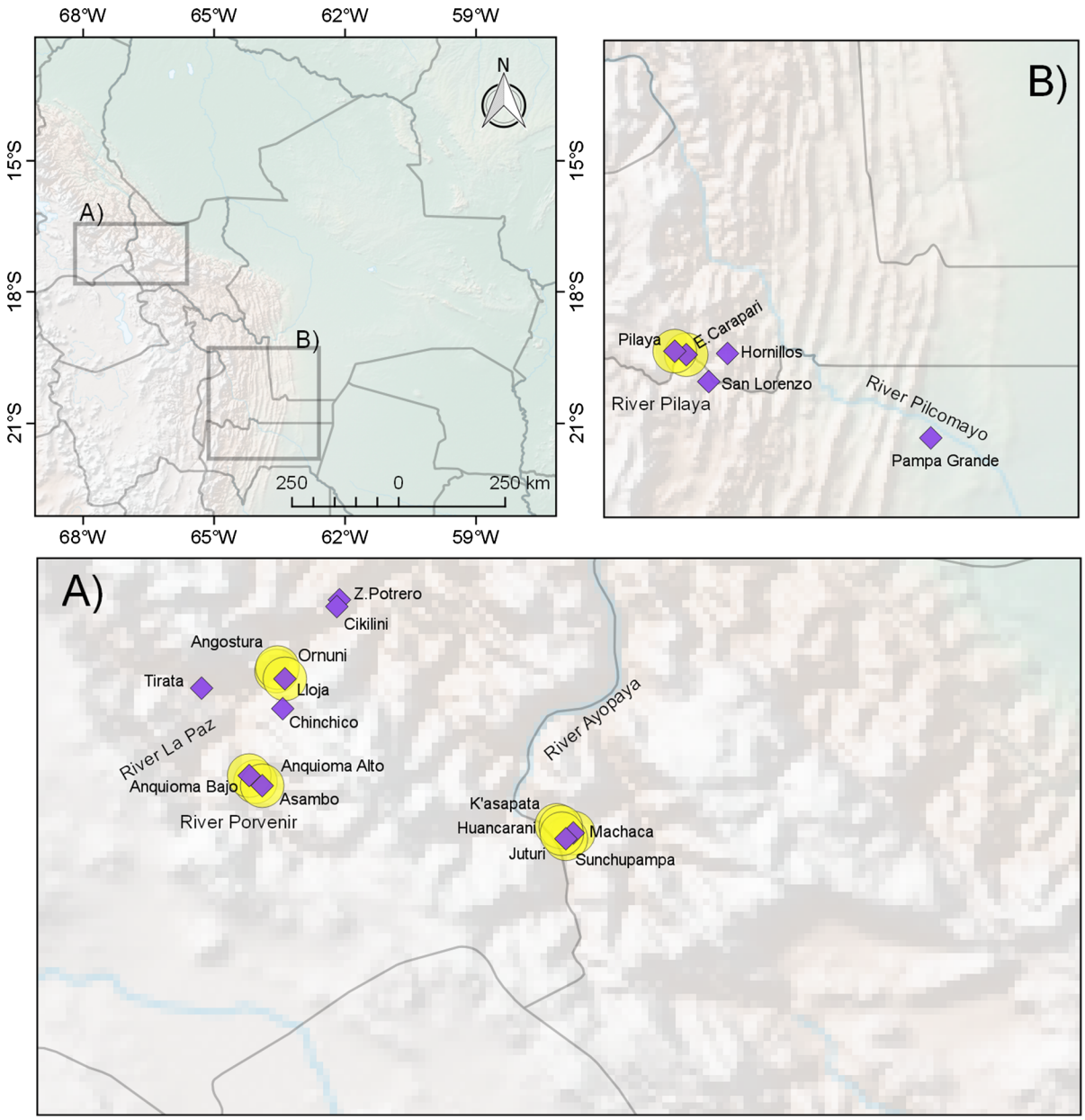

Fig. 1 **A–B** Overview of ahipa producer zones in Bolivia, and compared study areas (**A** and **B**) in years 1994/96–1999 (*diamond dots*) and 2012 (*circles*)

grew ahipa (*ajiperos*) in the 1990ties (Ørting 1994), i.e. in the communities Chinchico, Tirata, Cikilini, Zona Potrero in the department of La Paz reported by Ørting (1994), did no longer cultivate the crop. We successfully collected data in June 2012 when the water level of the rivers was low. In January to March it was impossible to visit the communities due to high water level and strong currents. In 2012, twenty-five farmers were contacted in Anquioma Alto, Anquioma Bajo and Asambo located in the Porvenir River basin. In the department of Cochabamba, field data were collected in January, February 1994 and April 1996 and again in 2012, i.e. in the rural communities of Machaca, Sunchupampa, Huancarani and K'asapata three farmers were contacted in 1994 (Ørting 1994) and nine in 2012 (Table 1). We also participated in

Table 1 Characteristic of study sites of ahipa producers in Bolivia

River Basin	Rural communities	Participants (N)		Gender				Growing (%)		No growing (%)		Average age of farmers		Planted area (ha)		Share of growers (%)	Change rate of planted area (%)
				Female (%)		Male (%)											
		1994/96	2012	1994/96	2012	1994/96	2012	1994/96	2012	1994/96	2012	1994/96	2012	1994/96	2012	2012	2012
La Paz	Angostura, Ornuni	0	8	0	0	0	1	0	11	0	0	n.a.	53	n.a.	0.18	n.a.	
	Lloja	6	4	0	0	6	4	6	3	0	61	45	64	0.17	0.06	1	-65
	Tirata, Cikilini, Zona Potrero	6	0	0	0	6	0	0	0	100	0	58	n.a.	0.04	0	n.a.	n.a.
Porvenir	Anquioma Alto, Anquioma Bajo, Asambo	8	47	4	11	4	36	8	49	0	44	27	45	0.54	0.64	8	19
Ayopaya	Machaca, Sunchumpampa, Huancarani, K'asapata, Juturi	6	17	4	6	2	9	6	5	0	44	35	58	0.03	0.006	1	−80
Pilaya	Estancia Carapari, Pilaya[a]	75	25	21	4	54	21	80	32	0	6	45	49	1.6	3.6	16	119
	Sample size (n)	**52**	**53**	15	11	37	41	49	37	3	16						

Bold values indicate at $p < 0.001$

Source: [a] Ørting (1996); Flores and Donaire (1999); Ponce-Tovar (2002)

field inspections organised by the municipal rural extension programme in May and June and collected information from the ahipa growing farmers. Ørting (1996) reported from interviews with seven farmers at the Estancia Caraparí in the Chuquisaca department; later Flores and Donaire (1999) reported information collected from 23 farmers in Pilaya. However, in 2012 only a total of thirteen could be contacted in both communities. We collected information in June and July 2012 when almost all farmers were completing the harvest of ahipa roots.

All interviews included both women and men, and we initially participated in a collective meeting to explain the research purpose. Farmers were consulted at the meeting on their willingness and time available to participate in the survey. We collected information about the farmers' field observations and in accordance with standard ethnobotanical methods conducted free listing and outdoor semi-structured interviews during crop management tasks, e.g. reproductive pruning, crop husbandry and root harvest (Gerique 2006; Hurrell and Pochettino 2014; Vogl et al. 2004). We followed the farmers while they carried out their daily tasks in their fields and took care not to disturb them excessively. Conversations during the participant observations were used to obtain information on traditional cropping systems, crop cultivation, seed management and the market price of roots of ahipa landraces. A modified semi-structured interview proposed by Ørting (Ørting et al. 1996) was used in 2012 to collect information and formed the basis for our comparison with the results obtained by Ørting et al. (1996).

Data analysis

Two diversity indices were estimated: i. The diversity of cultivated crop species managed by farmers at community level, and ii. The ahipa diversity at farmer and community level. Both indices were compared according to available data collected in 1994/96 and in 2012. The Brillouin index of diversity (*HB*) and diversity of cultivated crop species or richness (*S*) were used to compare communities and cropping periods. *HB* is considered suitable when sampling is non-randomized or complete. It is used for small collections (Magurran 1988; Peroni et al. 2014; Peroni and Hanazaki 2002; Pielou 1966):

$$HB = \frac{\ln N! - \sum \ln n_i!}{N} \quad (1)$$

where N is the total number of cultivated crops in the community, and n_i the number of cultivated crops grown by each farmer. The *HB*-value was used, considering that records of all crop species cultivated by each farmer are 'a collection' and not 'a sample'; the *HB*-value and cultivated crop species (*S*) within each of the farmer's plots were calculated for 1994, and for the Estancia Carapari with the data of 1996; for 2012 the information encompassed all communities. We compared *S* and to what extent farmers have replaced ahipa with other crops. We further assessed *S* and *HB* to estimate the agricultural diversity retained by *ajiperos* over the two study periods and for each rural community.

Richness is an intuitive definition drawn from the ecological point of view (Magurran 2004). A simple count due to few landrace types cultivated per household level, we applied the Margalef index. This index has zero as the lower limit when only one landrace is cultivated (Margalef 1958; Magurran 2004). It writes as:

$$MAD = \frac{(S_i - 1)}{LN(A_i)} \quad (2)$$

where *MAD* stands for the ahipa diversity maintained by a farmer ($MAD \geq 0$); A_i is the total area planted with all ahipa landraces; LN = natural logarithm, and S_i is the total number of ahipa landraces (Margalef 1958).

Statistical analyses were performed using the R software (R Core Team 2015) to assess significant difference in area, plant density, cycle length and root yield of ahipa landraces among rural communities. The non-parametric test of Mann–Whitney Wilcoxon was used to test whether the two cropping periods 1994/96 and 2012 were similar. Moreover, the new landraces recorded in 2012 were analysed by a one tailed *t* test of one sample. Georeferenced data were represented in a QGIS map (Quantum GIS 2015). Genetic Integrity (*GI*) and Erosion (*GE*) were estimated according to Hammer et al. (1996, 2005). Where, genetic integrity, GI = denotes the number of collected samples (landraces) of ahipa in the communities (*LR*) cultivated in 1994/96 and 2012, $GI = (LR_{1994/96}/LR_{2012}) \times 100$. Estimated genetic erosion was $GE = 100\% - GI$. *GE* rate was

determined on a yearly basis (%), considering a constant annual loss rate of genetic diversity per community (Hammer and Laghetti 2005). Using the formula the number of recorded landrace samples in 1994/96 and 2012 were compared. Likewise, in 2012, information on ahipa root prices from all communities was collected and compared with 1994/96. Commonly, ahipa root is tradeable at urban markets by a *Libra* (0.45 kg), and in producer communities by an *Arroba* (11.5 kg). In 2012, information on ahipa root prices from all communities was collected and compared with 1994/96. Besides the price of maize cobs—a.k.a. '*choclo*', potato and tomato in four communities were gathered and compared for two cropping periods. Prices in Bolivian currency (BOB) were converted to United States Dollar (USD) and compared to depict variations between the two cropping periods.

Results

Changes of agro-biodiversity and the status of ahipa in traditional farming systems

We present and compare results from the two periods, and discuss the effect of changes in agro-biodiversity on ahipa on-farm conservation in eleven rural communities located in the IAVs. The Brillouin diversity index (*HB)* and the crop diversity index or richness of crop species *(S)* were calculated for the communities located in each river basin. When (*HB*) values are higher than 1.0, they reflect more diversity among cultivated crop species, while (*S*) denotes higher overall crop diversity managed by farmers. *HB* and *S* are arranged from large to small values for comparing two periods (Fig. 2A–D). For the 2012 period *HB* ranked from 0.68 to 1.63, i.e. high, for the three communities, Angostura, Lloja and Ornuni in the La Paz River basin (Fig. 2A). The communities Lloja and Ornuni had high *S* in 2012 and 1994. Lloja and Cikilini had a low *HB* of 0.79 and 0.35 respectively. Lloja and Cikilini had *S* values of only 4 and 2 crop species, respectively.

In 1994/96 the recorded seed diversity in ahipa landraces was lower compared to 2012. We have only information about five landraces (b1a/wi1, xa3, b2 and rb) from four rural villages of La Paz River basin, one in the Porvenir River basin and two in the Ayopaya River basin, (Fig. 3A–C). The diversity of ahipa landraces (*MAD*) used by the *ajiperos* and the area planted did vary between the two cropping periods. The land used by the individual *ajipero* to cultivate ahipa does reach a quarter of hectare. The communities of the La Paz River basin use even less land to grow ahipa; only one landrace was grown in 2012 (Table 2). In the community of Lloja, two landraces were recorded in 2012. In this river basin the land used for growing ahipa was less compared to the remaining ahipa producing communities (Fig. 3A); i.e. the diversity was low (0.16) according to the Margalef Index.

In Angostura and Ornuni, traditional pre-Columbian terraces, known as '*Andenes*' in Spanish and '*Taqana*' in Quechua, were discovered (Fig. 4A). Several terraces still remained in good condition and were used for growing ahipa and green beans (*Phaseolus vulgaris* L.). *Pachyrhizus ahipa* were cultivated in the taqana (15 ± 5 m^2) associated with Lima beans (*Phaseolus lunatus* L.), and with achoccha (*Cyclanthera pedata* (L.) Schrad.) along the borders. Lloja is a community where the cultivation of cherimoya (*Anona cherimola* Mill.) has been popular for a long-time. Here the fragmented plots, known as '*Chiru*' (10 ± 3 m^2), are used for growing ahipa, tomato (*Lycopersicum esculentum* Mill.), bell pepper (*Capsicum annuum* L.), green beans, arracacha (*Arracacia xanthorrhiza* Bancr.), manioc (*Manihot esculenta* Crantz) and groundnut (*Arachis hypogaea* L.) as intercropping and used in the crop rotation.

The *HB* and *S* values in the communities of the Porvenir and Luribay River basins were lower. Anquioma Bajo had 1.76 of *HB* in 1994 that decreased to 1.69 in 2012 and *S* decreased to only nine crop species in 2012 (Fig. 2B). We did not estimate *MAD* and land use for 1994 due to lack of information. However, in 2012 mixed groups of coloured seed were recognized. Estimated *MAD* was higher (0.35) with cultivation in varied plot sizes (Table 2; Fig. 3B). The *P. ahipa* and the popular marketable crops were grown in small plots—a.k.a. '*K'ata*' (50 ± 8 m^2); furthermore, tomatoes or ahipas are often intercropped with maize in one or two '*K'atas*' (Fig. 4C). Arable land is limited and the crop association with traditional cropping systems is still practiced by farmers. Ørting (1994) described the cultivation of larger plots with ahipa, the land area unit known as '*marka*' (200–400 m^2) was reported. However, at present many *ajiperos* grow cut flowers, white maize cobs and tomato crops in small split plots.

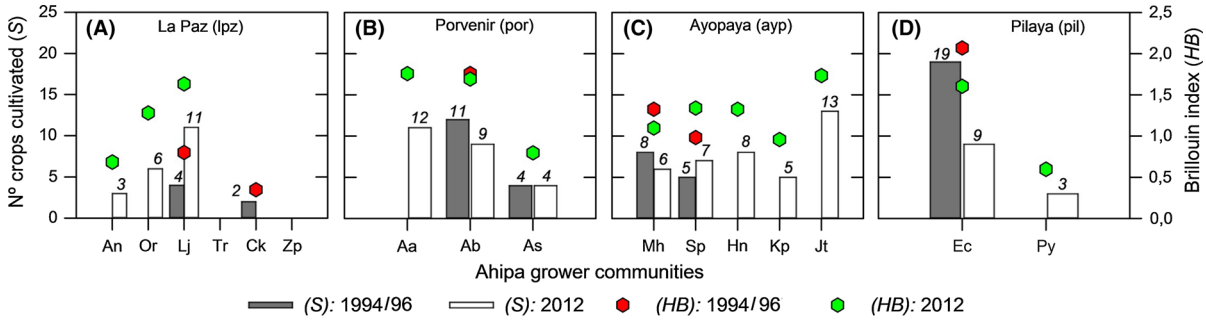

Fig. 2 **A**–**D** Richness and Brillouin diversity index of crop species cultivated. Number in Italic over the *bars* show the number of crop species recorded in 1994 and 2012 with exception to Estancia Caraparí (Ec) for 1996 year. Ahipa rural community producer **A** La Paz: Angostura (An), Ornuni (Or), Lloja (Lj), Tirata (Tr), Cikilini (Ck), Zona Potrero (Zp), **B** Porvenir: Anquioma Alto (Aa), Anquioma Bajo (Ab) and Asambo (As), **C** Ayopaya: Machaca (Mh), Sunchupampa (Sp), Huancarani (Hn), K'asapata (Kp), and Juturi (Jt), and **D** Pilaya: Estancia Caraparí (Ec), Pilaya (Py)

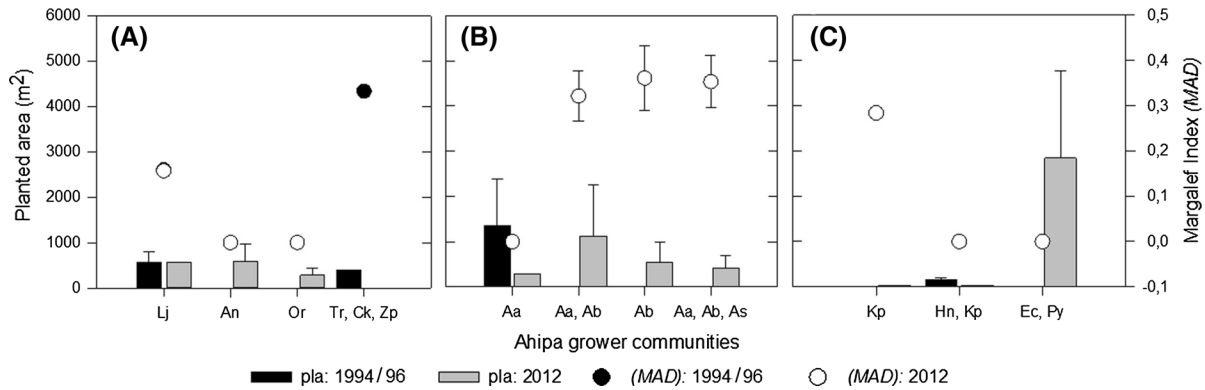

Fig. 3 **A**–**C** Margalef ahipa diversity index *(MAD)* of cultivated ahipa landraces diversity at community level and planted area (pla) in 1994 and 2012 with the exception of the Estancia Caraparí (Ec) and Pilaya (Py) for 1996. Rural river basin communities—ahipa producers: **A** La Paz (lpz), **B** Provenir (por), and **C** Ayopaya (ayp) and Pilaya (pil). Italic number over the *bars* show the number of ahipa landraces recorded. *Error bars* denote the standard deviation of estimated *MAD*

The community of Machaca in the Ayopaya River Basin was recorded with a high diversity (*S*) in marketable crop species. *HB* was 1.33 in 1994 and decreased to 1.10 in 2012, whereas diversity richness (*S*) was 8 in 1994 and had in 2012 declined to 6 (Fig. 2C). Mixed cropping systems were observed in Juturi, K'asapata, Huancarani and Sunchupampa all in the Ayopaya River basin. Farmers have small plots combined with fruit orchards, e.g. cherimoya, avocado (*Persea americana* Mill.), 'creole orange' (*Citrus* sp.), 'creole lime fruit' (*Citrus* sp.) and papaya (*Carica papaya* L.). Plots called *'ananta'* (63 ± 12 m^2) in Quechua were used for growing tomato, maize, locoto (*Capsicum pubescens* Ruíz et Pav.), sweet potato (*Ipomoea batatas* L.) and potato (*Solanum* spp.) as well as ahipa. In 2012 the estimated diversity (*MAD*) was 0.28 for ahipa cultivated in reduced plot size (Fig. 3C).

In 1996 the number of crop species was scored for a single community in the Pilaya River basin only, i.e. the Estancia Caraparí with 19 crops and with a higher Brillouin diversity index (2.07) than other communities (Fig. 2D). However, in 2012, *HB* was lowest for the Estancia Caraparí (1.6) and Pilaya (0.6); here an ahipa landrace is an economically attractive, seasonal food crop when marketed during the Corpus Cristi festivity (*MAD*: 0). It is often grown in large plots called *'parcela'*(2840 ± 1930 m^2, see Fig. 4E).

Agronomic changes in two cropping periods

We compared planted area, plant density, cycle length and root yield of two cropping periods 1994/96 and 2012. The growth cycle of ahipa landraces differed among rural communities, but not between cropping

Table 2 Accounted cultivated specie crops and ahipa landraces in the localities

River Basin	Rural Community	Altitude (m)	Diversity of cultivated crop species										
			N		S		r		*E*		Ahipa (n)		
			1994/96	2012	1994/96	2012	1994/96	2012	1994/96	2012	1994/96	2012	Landraces
La Paz	Angostura	1733	–	5	–	3	–	2	–	0.83	0	1	w1/b1
	Ornuni	1691	–	11	–	6	–	5	–	0.84	0	1	w1/b1
	Lloja	2400	4	13	4	11	0	2	1.28	0.97	2	2	w1a, b1a
	Tirata	2200	–	–	–	–	–	–	–	–	3	0	xa3
	Cikilini	1712	2	–	2	–	–	–	0.00	–	3	0	xa3
	Z. Potrero	1973	–	–	–	–	–	–	–	–	3	0	xa3
Porvenir	Anquioma Alto	2366	–	46	–	11	–	35	–	0.40	1	3	b2, xb1, xb2, xb3
	Anquioma Bajo	2368	17	38	12	9	5	29	0.91	0.40	1	2	b2, xb1, xb2, xb3
	Azambo	2463	4	4	4	4	0	0	1.28	1.28	1	1	b2, xb1, xb2, xb3
Ayopaya	Machaca	2759	8	6	8	6	0	0	1.07	1.12	1	1	sb, rb
	Sunchupampa	2058	6	11	5	7	1	4	1.00	0.88	1	1	sb, rb
	Huancarani	2146	–	8	–	8	–	0	–	1.07	0	1	sb, rb
	K'asapata	2191	–	5	–	5	–	0	–	1.17	0	2	sb, rb
	Juturi	1972	–	13	–	13	–	0		1.03	0	1	sb, rb
Pilaya	Azurduy		5	–	5	–	0	–	1.17	–	0	0	
	E. Carapari	1172	–	37	–	9	–	1	–	0.86	2	1	b3
	Pilaya	1312	–	3	–	3	–	0	–	1.63	2	1	b3

Collected information on cultivated crop species from farmers in the localities, where: total number of crops in the sample (N), species number of crop cultivated (S). Variables as richness (r), and evenness or equitability (E) were calculated by Brillouin diversity index

periods ($p = 0.421$). Growth cycle period differed significantly ($p < 0.05$) among landraces in 1994/96 and 2012. Landraces [rb] and [b2] from the Ayopaya and Porvenir River basins were grown in 277 and 262 days in 1994/96, and 242 and 302 days in 2012 (Fig. 5A). In the Lloja community of the La Paz River basin, landraces [b1a] and [w1a] had a growth period of 211 days in 1994/96 and 166 days in 2012. In the communities of Angostura and Ornuni of the La Paz River basin, two landraces were cultivated [b1] and [w1] with a growth cycle duration of 166 and 182 days in 2012 (Fig. 5A). We collected seed and described the colour of the testa. In 2012, farmers cultivated mixed landraces of: [xb1], [xb2] and [xb3] with growth cycles of 266, 277 and 261 days (Fig. 5A) in the rural river basin communities of the Porvenir River basin (Anquioma Alto, Anquioma Bajo and Asambo). Additionally, in 2012 a growth cycle of 288 days was recorded for a landrace [sb] of the Ayopaya River basin (Machaca, Sunchupampa, Huancarani, K'asapata and Juturi communities).

Plant density per square meter was not significantly different ($p = 0.40$) for the 1994/96 and 2012 and landraces ($p = 0.50$). However, density differed between the new collected landraces in 2012 ($p = 0.008$). For the landraces [b1a/w1a], [b2] and [rb] the plant densities were 9, 11, and 7 pl m^{-2} in 1994/96 in the river basin communities of La Paz, Porvenir and Ayopaya. In the communities of the Pilaya River basin, the landrace [b3] had a high plant density (16 pl m^{-2}) in 1994/96, and 23 pl m^{-2} in 2012. Plant densities for the landraces [b1a/w1a], [b2] and [rb] were 19, 20 and 14 pl m^{-2} in 2012, respectively (Fig. 5B). An estimated plant density for the

A
B
C
D
E

◀**Fig. 4** **A–E** Seed, roots of ahipa landraces and traditional cropping systems in River Basin communities of Bolivia. **A** w1/b1; *Taqana*, La Paz (lpz), **B** w1a, b1a; *Chiru*, La Paz (lpz), **C** b2, xb1, xb2, xb3; *K'ata*, Porvenir (por), **D** sb, rb, *Ananta*, Ayopaya (ayp), **E** b3, *Parcela*, Pilaya (pil), photos J.P. Rodriguez

landrace [xa3] in 1994/96 was 6 pl m^{-2}, but observations were not available for 2012.

Root yields were not significantly different ($p = 0.181$) between the two cropping periods (1994/96 and 2012), nor among the landraces ($p = 0.228$). Nevertheless, there were significant differences among the new landraces identified in 2012 ($p < 0.001$). In 1994/96 the landraces [b1a/w1a], [b2], [rb] and [xa3] produced yields of 47, 42, 39.8 and 40 t ha^{-1}. The communities of the Pilaya River basin with the landrace [b3] had a higher root yield in 2012 (66.1 t ha^{-1}). Also in 2012 the landraces [rb] and [b1a/w1a] yielded 40.5 and 39 t ha^{-1}, respectively (Fig. 5C). Conversely, the landrace [b2] had a low yield the same period (20.2 t ha^{-1}). No information was collected for the landraces [b3] in 1994/96 or for [xa3] in 2012.

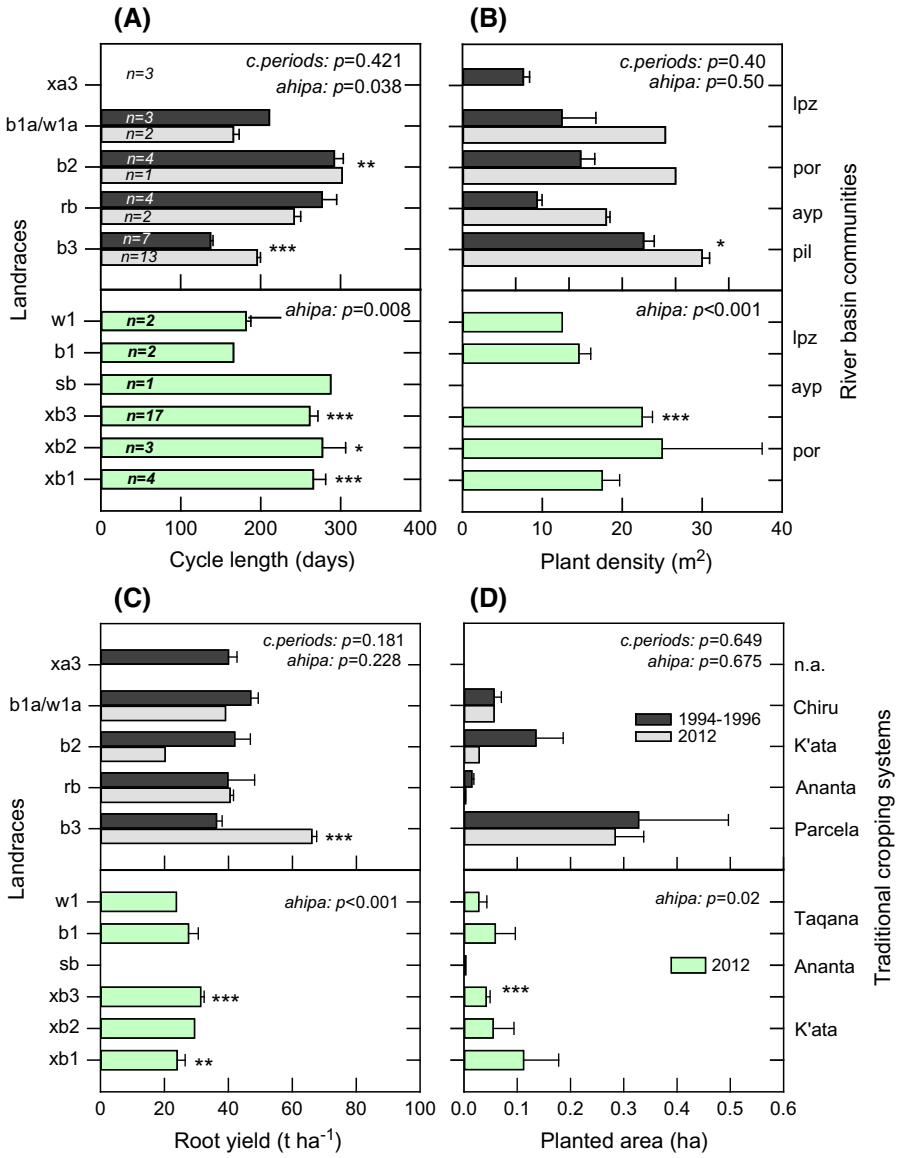

Fig. 5 **A–D** In-situ comparisons of ahipa landraces in two periods 1994/96 (*bars* in *black colour*) and 2012 (*bars* in *gray colour*): **A** cycle length, **B** plant density, **C** root yield and **D** planted area. Ahipa landraces were defined by their testa colour. Subfigures in lower half of each of the four graphs correspond to new ahipa landraces identified in 2012 (*bars* in *green colour*). Abbreviations explained in Fig. 3. (***) and (**) Significant differences between landraces at $p < 0.01$ and $p < 0.001$ and (*) to $p < 0.05$ under *t* test

There were no differences between 1994/96 and 2012 in area planted with ahipa ($p = 0.649$) nor among landraces ($p = 0.675$). The planted areas differed significantly between the new landraces in 2012 ($p = 0.02$). In 1994/96 the planted area was higher (Fig. 5D) for landraces [b1a/w1a], [b2] and [rb] in the communities of the La Paz, Porvenir, and Ayopaya River basins. Ahipa landraces [b3] in the Pilaya River basin were planted on a larger area (0.328 ha) in 1994/96 compared to 2012 (0.284 ha). In contrast, landraces [b1a/w1a] and [b2] were cultivated on smaller plots in 2012 (Fig. 5D). A reduced area of 0.003 ha was estimated for the cultivation of landrace [rb], and we were unable to estimate the area planted with landrace [xa3] in both cropping periods 1994/96 and 2012 due to information being unavailable and inaccessibility to the community.

Genetic erosion (*GE*) between 1994/96 and 2012 was estimated; the calculated loss of ahipa landraces was 67% for the [b1a/w1a] from Lloja (in the La Paz River basin), for [b2] from Anquioma Alto (in the Porvenir River basin) and 48% for [b3] from the Estancia Carapari and Pilaya. The current genetic integrity (*GI*) was 33% for both [b1a/w1a] and [b2]. The ratio is calculated according to the currently available landraces per community in quantity by the number of landraces per community grown in Lloja (La Paz River basin) and Anquioma Alto (Porvenir River basin). For Estancia Caraparí and Pilaya (Pilaya River basin) it corresponds to 52% loss of landraces.

Socio-economic factors and landrace diversity

The number of ahipa farmers per rural community comprised between 1.0–16.0% of the total local population in 2012 (Table 1), however, this calculation cannot be completed for the 1994/96 study due to lack of records. Albeit, the two surveys do show that the Bolivian Andean yam bean has experienced changes in crop husbandry and root production. Crop farming is the main activity in the communities of La Paz River basin; the *Ajiperos* account for approx. 1% of the total household population of the Angostura, Ornuni and Lloja communities, i.e. communities that have continued to grow *P. ahipa* according to traditional techniques. Land is generally inherited, and property sizes range from 0.03 to 0.06 ha (Table 1). The vernacular denominations of seed in the Aymara and Quechua dialects are shown in Table 2.

The proportion of ahipa growers declined by 44–61% from 1994/96 to 2012 in the La Paz, Porvenir, and Ayopaya River basin communities. In 1994 the youngest *ajiperos* were on average 27 years in the Anquioma Alto, Anquioma Bajo, and Asambo communities (in the Porvenir River basin) and 45 years in 2012 (Table 1). Ahipa growers constitute 8% of the total population of the Porvenir River basin communities. Smallhold sizes range from 0.03 to 0.11 ha, and crop farming relies on popular cash crops followed by seasonal cut flowers and ahipa. Men were involved with crop caring and harvest, whereas women were mainly involved in post-harvest handling. Negative rates of the area planted with ahipa were reduced by −65% in the La Paz River basin and by −80% in the Ayopaya River basin from 1994/96 to 2012 (Fig. 5D). The communities of Machaca, Sunchupampa, Huancarani and K'asapata in the Ayopaya River basin are characterised by diversified farming systems with small farm sizes (average 0.02 ha) having high agro-biodiversity richness (Fig. 3C). The ajiperos, aged 35 years (in 1994) and 58 years (in 2012), account for 1% of the total households in the communities. Traditional farming systems are based on popular cash crops, Andean root and tubers, onion, fruit orchards (e.g. cherimoya and avocado) and a little area with ahipa.

The number of Ahipa growers from the communities in the Pilaya River basin communities accounted for 80% of the farmers in 1994/96, but declined to 32% in 2012; this is in contrast to the findings in La Paz River basin communities where the cultivation was less in 1994/96 but did show a marginal increase in 2012 (Table 1). Overall the *ajiperos* account for 16% of the total households of the Pilaya and the Estancia Carapari communities. Their average age varied only marginally between the two surveys, i.e. from 45 to 50 years. The local farmers grow sugar cane (*Saccharum officinarum* L.), potato and other popular cash crops. Farm size was 0.28 ha on average, which is larger than in the other ahipa producing communities. 54% of the farmers contacted during crop husbandry tasks were men past middle-aged. Only 21% of the interviewees were young females engaged in reproductive pruning and post-harvest tasks.

In 2012, when we visited the communities of the river basins of La Paz (Lloja), Porvenir (Anquioma Alto, Anquioma Bajo, Asambo) and Ayopaya (K'asapata, Huancarani, Machaca, Sunchupampa and

Juturi), the number of *ajiperos* had become reduced by 61 and 44%, respectively (see Table 1). The number of agriculturalists devoted to ahipa cultivation was on average 6% of the total population per rural community. Ahipa growers, in the La Paz and Ayopaya River basin communities were in 1994/96 found to be between 27 and 58 years old whereas in 2012 their age had increased to 45–64 years (Table 1). Furthermore, in the same rural communities the area planted with ahipa had been reduced by −65% (La Paz River basin) and by −80% (Ayopaya River basin). However, in 2012 the communities of the Pilaya River basin had extended the area with ahipa compared to 1994/96 (Fig. 3C).

Seed and ahipa root survival at on-farm situation and marketing

Farmers in the communities of the La Paz River basin used and conserve seeds of separated colours. In 2012, we described the testa colour and root morphology of 12 landraces (Table 3). The ahipa roots have sweet white flesh with purple stripes and with two root shapes: obovate and oblong. In the Angostura and Ornuni communities of the La Paz River basin, the white-yellowish and black mottled testa colour are branded as "*ch'eje*" (Table 3; Fig. 4A), an Aymara word. Two farmers grew ahipa in 2012 in each community of Angostura and Ornuni, but only one existed in Lloja, i.e. also in the La Paz River basin (Fig. 4B). In 2012 we often met farmers busy with root harvest and women engaged in the collection and selection of seed pods. In the Porvenir River basin communities, farmers conserve seeds with mixed colours. Large seed size is preferred when selecting the best genotypes (Fig. 4C). In 2012, we met two farmers who did separate seeds into two colours, i.e. dull black, known as café, and white-yellowish mottled black (Table 3). Two types of root flesh colour are featured: 1. white with a narrow circle of purple stripes and oblong in shape, and 2. white without stripes and with an ovate shape. In the communities of Ayopaya River basin (K'asapata, Sunchupampa, Huancarani and Juturi), two different testa colours were identified (Fig. 4D) and root with variations of flesh sweetness and shapes: a) shiny black seeds called *"yana"* (Quechua), and obovate sweet fleshed root and b) reddish brown seeds known as *"chumpi"* with a non-sweet fleshed, oblong root. In 2012, only two farmers in the Ayopaya River basin communities of K'asapata and Huancarani communities grew ahipa (Table 3), one had only small, not yet harvestable roots. At Estancia Caraparí and in the community of Pilaya both in the Pilaya River basin only the dull black seed colour (Fig. 4E) is used; known to the farmers as '*negrita*', i.e. black coloured seeds. The root has an obovate shape and sweet flesh with a narrow ring of purple dots (Fig. 4E). Many male farmers were busy packing the harvested roots, while the women and children gathered the mature pods. During and after root harvest the women were responsible for selecting roots for the market prior to the packing.

Ahipa roots are sold at urban markets in the cities of La Paz, El Alto, Sucre, Cochabamba, and Tarija. In Bolivia, the crop production relies exclusively on the cultivation of landraces. Commonly ahipa is traded by local units a.k.a. *libra* (0.45 kg) and arroba (11.5 kg). The price obtained for roots of ahipa landraces changed over time (Fig. 6). Over the two periods the ahipa from the La Paz, Porvenir and Pilaya River basin communities, experienced an increase in price of 125%. While the market price in the Ayopaya River basin communities –decreased 100% (Fig. 6). Here commercialization of ahipa roots between rural communities is rare and the crop is included in the crop rotation and intercropped systems (Fig. 5A, D). Crop species belonging to the Solanaceae, Fabaceae and Poaceae are more frequently cultivated than ahipa. Crops like potatoes, tomatoes and white maize cobs are more frequently traded than ahipa. The prices increase more for white maize cobs and potatoes than for ahipa roots from 1994/96 to 2012 (Fig. 6). The *ajiperos* often mentioned that maize has a short growth cycle and cobs may be harvested up to even three times annually. If combined with a commercial crop in a rotation system, maize can still be grown twice a year. Similarly, tomatoes provide more frequent harvests than ahipa, once the plants have been established. The fruits—like white maize cobs—can also be collected several times per year. In 2012, white maize cobs and potatoes achieved a 200% higher price per *arroba* than ahipa compared to 1994/96 (Fig. 6). Albeit, with the optimal timing of the planting the ahipa roots can attain prices similar to that of maize cobs and potatoes per *arroba* or even a little higher. However, the ahipa growth cycle is longer (between 6 and 8 months). In the southern region, the Pilaya River basin, the price ahipa did did increase 23%.

Table 3 Seed and root features of identified ahipa landraces

Landrace	Seed testa—colour feature	Seed shape	Flower colour	Root flesh colour	Root shape
w1	Yellowish white/black—mottled *("ch'eje")*	Rounded kidney	White	White flesh with purple stripes	Obovate
b1	Dull black	Rounded kidney	Purple		Oblong
w1a	Yellowish white/black—mottled	Rounded kidney	White		
b1a	Dull black	Rounded kidney	Purple (locally called *"azul"*, blue)		
xa3	Dull black/purple yellowish mottled black/white yellowish mottled black (mixed group of seeds)	Rounded kidney	Two flowers:colour: blue and white		
b2	Dull black, rarely known as *"café"*	Rounded kidney		White flesh with a purple striped narrow ring below the epidermis	Oblong (*"ch'acuro"*)
	White yellowish mottled black	Rounded kidney	WHITE	White flesh without purple stripes	Obovate
	Purple yellowish mottled black				
	*Farmers hold three mixed colour combinations of seed:				
xb1*	Dull black/white yellowish mottled black,				
xb2*	Dull black/purple yellowish mottled black,				
xb3*	and dull black/white yellowish mottled black/purple yellowish mottled black				
sb	Shiny black *("yana")*	broad kidney	purple and white	white flesh with purple stripe	obovate
rb	Maroon/reddish brown *("chumpi")*	broad kidney	purple	white flesh without stripes	oblong
b3	Dull black *("negrita")*	rounded kidney	purple and (rarely) white	white flesh with a narrow ring of purple spots below the epidermis	obovate

Discussion

Trends in consumer demands and pressure from marketable crops on the agro-biodiversity of ahipa landraces

Our study in Bolivia compared the in situ conservation of the Andean yam bean associated with traditional farming systems involving the ethno-ecological knowledge use and change of agro-biodiversity over two cropping periods, i.e. 1994/96 and 2012. The ethno-ecological knowledge contributed significantly to the on-farm conservation of the crop. The in situ monitoring demonstrated that farmers had organized ahipa cultivation according to traditional farming systems (Figs. 4A–E, 5A–D). However, the cultivation of marketable vegetables, legume, tuber and grain crops had increased quantitatively from 1994/96 to 2012 when a wide range of conventional crops was grown dominated by species belonging to the Solanaceae and Fabaceae. Initiatives of in situ seed conservation are required to ensure ahipa from being neglected or even becoming extinct. An example: a decrease in ahipa cultivation took place in communities of the Porvenir River basin, where Ørting (1994) recorded large plots known as *'marka'*, whilst *ajiperos*

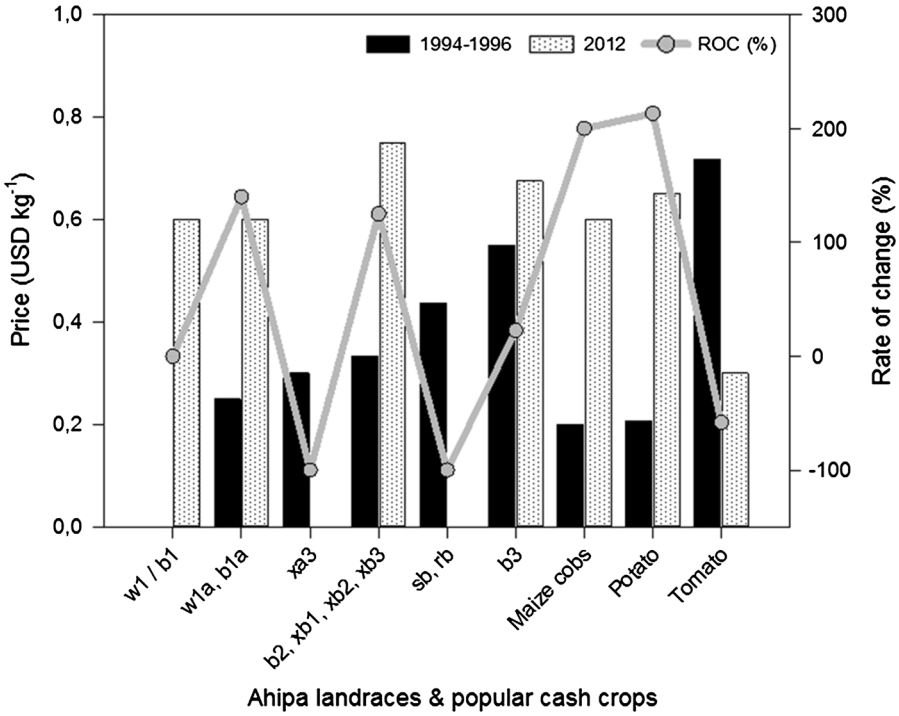

Fig. 6 Price of ahipa and popular rentable crops in 1994/96 (USD 1.0 = BOB 4.6) and 2012 (USD 1.0 = BOB 6.96) (Source: https://www.bcb.gob.bo/tiposDeCambioHistorico/), ROC: Rate of change in percent (%)

divided their plots into smaller units called *'k'ata'* in 2012. Regarding the decline in the cultivation of ahipa, two important factors shall be taken into account: the laborious reproductive pruning, i.e. to ensure the translocation of assimilates to the tuberous root rather than seed growth, and the loss of seeds due to been weevils/bruchids. Both factors might be eased in further collaboration between peasants and agronomists; helping to get a deeper understanding of when the reproductive pruning is necessary and most effective, and easing the trading/exchange of seeds between villages and regions.

In most of the communities visited, the farmers belonged to the elder generation were conscious of the importance ensuring the conservation of agro-biodiversity. Ahipa remains a particular case for which no conservation measures have so far been carried out whether in situ or ex situ. In-situ seed conservation initiatives in combination with the traditional knowledge of crop diversity are needed to preserve ahipa from being neglected or even becoming extinct. The *ajiperos* can be regarded as *'biodiversity custodians'* (Paudel et al. 2012; Gruber et al. 2013, Galluzzi and López-Noriega 2014a; Jarvis et al. 2011). Traditional farming systems, as e.g. *chiru, ananta* and *k'ata*, were dynamic units as production areas in both time periods (Fig. 2A–C). These traditional farmiing systems enabled farmers to maintain a high agro-biodiversity including marketable crops similar to the ones described by Zimmerer (1992, 1996, 2013) and Zimmerer et al. (2017) in Paucartambo, Peru and the IAV communities, Bolivia. This high agro-biodiversity cultivated on small land sizes (Fig. 5D) is competing with ahipa, and we found that ahipa landraces are almost exclusively conserved by older farmers (Table 1). Meanwhile current farmers are practising intensive agriculture involving marketable crops (Table 3). The verticality of agriculture in these tradtional farming systems allowed the conservation of ahipa, however, crop boundaries can move when humans provide special attention (Gade 2016). Priority is increasingly given to short growth-cycle cash crops that are less labour demanding, e.g. the Poaceae (cereals) and the Solanaceae (vegetables). An exception is that eighteen ahipa seed accessions are presently preserved by the National Research Institute of Agriculture and Forestry (INIAF) (Terrazas and Gonzáles 2011; Galluzzi and López-Noriega 2014a, b); albeit, their ex situ conservation status remains unknown. In-situ seed conservation is poorly managed without technical support in remote rural communities, i.e. Lloja, Angostura and Ornuni in the

La Paz River basin. Farmers conserve the germplasm based on seed colour and size as observed in the Porvenir River basin communities without further selection processes (Table 2; Fig. 4C). If the crop diversity becomes low, this decline is caused by the historical lack of link between peasants and markets (Soluri 2013). Rodriguez et al. (2017) pointed out that remoteness of communities in the IAV is the main reason for the lack of contact with urban markets.

In the Lloja community, ahipa diversity remained unchanged for the two periods studied. The two landraces reported in 1994/96 and 2012 are conserved by one grower only, but were in 1994/96 reported cultivated by several farmers (Fig. 3A). Thus, as Brown and Hodgking (2015) stressed one of the factors of genetic erosion is the loss of traditional knowledge and experience among local farmers. Therefore, it seems that ahipa diversity in this river basin would have this loss whether or not conservation measures are taking into account. Seed exchange of ahipa occured only within Lloja, hence at present, this single farmer has been forced to use his own seeds exclusively. The agricultural land is restricted and vertical agriculture is practiced and has even been reestablished here before the communities of Angostura and Ornuni. The transportation for people is very cumbersome for bringing agricultural products to the nearest bus station, i.e. 3 h or more to the nearest community (Rodriguez et al. 2017). The Bruillon index estimated higher diversity of crop species in 2012 than in 1994/96, genetic diversity was highly dynamic (Table 2). Marketable crops with short growth cycles are grown, and here traditional farming systems were dynamic and not static (Fig. 2A). Agriculturalists have wisely managed and increased the crop diversity considerably.

Ahipa seed mixtures were observed in the communities of the Porvenir River basin, involving three differently coloured ahipa landraces. However, farmers do not pay attention to the time of harvesting and the preparation for selling at the market. Here, many farmers have wisely diversified the small land units through crop rotating, poly-cultures and combined cropping systems. In 1994/96 large unit area ahipa farming was reported from the Anquioma Alto locality by Ørting (1996), but in 2012, the area with ahipa has become smaller (Fig. 3). Similar processes take place in the other localities studied where the crop diversity similarly is dynamic (Fig. 2B).

Although, the communities K'asapata, Huancarani, Sunchupampa, and Juturi are located far from Independencia rural municipality in the Ayopoya River basin (Figs. 1A, 4D), the presence of two ahipa landraces with differentiated root types marked an important diversity. Sadly, nowadays the ahipa farmers have reduced land available, on sloping ground, compared to the rest of cultivated lands, here ahipa is left to its own fate. Marketable crops are increasingly grown (Fig. 3C). Only a few crops were reported in 1994/96, but in 2012 more crops were planted in most of the communities (Fig. 2C); they have gradually increased in popularity at the benefit of the ahipa.

The change in genetic erosion (GE) observed between 1994/96 and 2012 for ahipa landraces [b1a/w1a] and [b2] was 67% in the Lloja and Anquioma Alto communities, and 48% in landrace [b3] in the communities of Estancia Carapari and Pilaya. The loss of landraces in those communities are similar in that fewer farmers are now interested in growing ahipa. As a result of this, only a single farmer produced ahipa in Lloja and in Anquioma Alto, where increased diversity in marketable and high yielding crop varieties was grown. Genetic integrity (GI) was 33% for landraces [b1a/w1a] and [b2]. This estimation corresponds to the agro-biodiversity—according to the available landraces per community—cultivated in Lloja (La Paz River basin) and Anquioma Alto (Porvenir River basin). This is the first attempt to compare the diversity in landraces according to data collected over two cropping periods two decades apart. Estancia Caraparí and Pilaya communities denote a loss of 52% in some landraces; in both communities, farmers used seed material of only one seed colour. As Brown (2008) and Brown and Hodgking (2015) stressed regarding the vulnerability of genetic diversity: the ahipa is both environmentally and spatially endangered in the communities mentioned above.

The effect of traditional agronomic practices and socio-economic factors on ahipa conservation

Agriculture in the IAV communities of Bolivia is delineated by significant environmental constraints; on this background, the study is a first comparison of on-farm conservation of *P. ahipa* by the *ajiperos*. Our agronomic and diversity quantification approaches revealed strong shifts towards popular marketable crops in the traditional cropping systems in

preference to the ahipa. In Bolivia this is a concern at the crop diversity level and as the country is the centre of origin of the ahipa as an orphan crop (Weddell 1857; Cardenas 1989; Sørensen et al. 1997). None-the-less the in-field plant density recorded in ahipa has increased over the past two decades (Fig. 5B); however, the ahipa root yield did not increase over time in all communities where similar yields are recorded even of newly identified landraces (Fig. 5C). At in situ conditions in the Pilaya River basin had the highest root yield compared to experimental conditions; Ponce-Tovar (2002) reported 9.8–13.8 t ha^{-1} in local ahipa accessions. However, in 1996 our study estimated yields at 38 t ha^{-1}, and in 2012 the yield doubled. Flores (2001) reported from field experiment in the SE department of Tarija in the Pilaya River basin communities root yields of 11.4–35.4 t ha^{-1}. Communities like the Estancia Carapari and Pilaya had higher root yields than the yields of other Bolivian landraces (Fig. 5C). In conditions of introduction (ex situ) of ahipa as a new crop in African countries, Grüneberg (2016) reported contrasting root yields: in Rwanda of 26.9 t ha^{-1}, two locations in the Democratic Republic of Congo 6.4–18.5 t ha^{-1} in Mulunga and 5–15 t ha^{-1} in the Ruzizi Valley, and finally in Burundi 19.3–23.2 t ha^{-1}. High plant density (Fig. 5B) resulted in low root yield in 2012 (Fig. 5C) for the ahipa landraces as a result of low soil fertility. Similar results were found by Høgh-Jensen et al. (2008). Agricultural intensification has according to Vandermeer et al. (1998), Jackson et al. (2007) and Junqueira et al. (2016) resulted in a declining crop diversity. Information on soil fertility is necessary, but due to logistical constraints such analyses were not conducted and we only estimated mean root yields and plant densities in ahipa based on information from local farmers and own observations recorded in 2012. Tittonell and Giller (2015) suggested that a significant fraction of the yield gap could be avoided by proper agronomic management. The introduction of short vegetative cycle crops is common and endangers the prospects of the ahipa. White maize cobs, tomato and potato have taken cultivation priority. Prices of white maize cobs did change compared to the price of ahipa (Fig. 6) as maize, potato and tomato have become profitable products in urban markets and rural communities. There is an ominous lack of opportunities regarding the promotion of ahipa in urban and new niche markets. Currently farmers are opting for monocrops and crops with short growth cycles (Khoury et al. 2014). A similar development can be seen in the rural IAV communities where ahipa is grown. Farmers rely on high seed variety of marketable crops from which seeds easily can be obtained at shops in all the cities.

The distance of the remote rural IAV communities from the urban centres has a high impact on the population dynamics (Cárdenas 1966; Rea 2004; Ørting 1996). According to Ørting (1996) young ahipa growers could still be contacted and interviewed in 1994/96 (Table 1), but at present, the major part of the younger generations have migrated to the cities, and most ahipa growers are more than 50 years old. This situation will accelerate the loss of traditional knowledge related to ahipa. To recover and conserve this knowledge it is necessary to develop and promote ahipa-based foods and organic farming (Segovia 2004; Doporto et al. 2011; Ramos-de-la-Peña et al. 2013). While at other latitudes, valuable research highlights the attractive qualities of ahipa (e.g. Dini et al. 2013; Doporto et al. 2011, 2014; Ramos-de-la-Peña et al. 2013; Doporto 2014). Unfortunately the species remains an orphan crop in Bolivia.

Conclusion

Traditional knowledge is of considerable value when it supports the conservation of rural agro-biodiversity. However, when lack of interest and support prevail, a native leguminous crop like *P. ahipa* is at risk of being lost. Changes in cultivation of traditional crops took place over a period of nearly 20 years, and it is clear that short seasoned, marketable crops are displacing the traditional crop ahipa. Low market price, agricultural extension, no collection at the national level and lack of seed are factors causing loss of interest to continue with ahipa cultivation in the La Paz and Cochabamba departments. Besides, when considering that ahipa is a rainfed crop, climatic conditions constitute a critical factor in the locations where the crop is currently cultivated. Also, as priority given to marketable crops that require less labour the prospects for the ahipa appear grim. Recollection of seed should be conducted in remote communities and ex situ conservation in seed banks should be implemented. Elderly ahipa growers should be valued for their local knowledge. Successful experiences with ahipa in the

department of Tarija should be promoted and exchanged among farmers from other communities. Faculties of Agronomy and research institutions should be encouraged to promote the food and nutritional qualities of ahipa to the Bolivian citizens. Regrettably it remains a matter of considerable concern that although 2016 was celebrated as the International Year of the Legumes, orphan legume crops like e.g. the Andean Yam Bean or other neglected leguminous crops were not promoted. The ahipa, as a potentially important endemic genetic resource with numerous attractive traits has yet to be promoted—failing this it may soon be too late to harvest the efforts made by generations of local Andean farmers.

Another aspect of concern is that Bolivia has yet to ratify the Nagoya Protocol, because the Bolivian Genetic Resources cannot be monitored if funds are not made available for this activity. Until this situation is altered Bolivia cannot exchange and cannot allow the introduction of foreign genetic material.

Acknowledgements We greatly appreciate the co-operation and participation of all farmers encountered in the rural communities, who were involved in the survey and shared their valuable knowledge on ahipa management and uses. We would also like to thank the local technicians and colleagues for the logistic support during the fieldwork to: Insfranz Leon, Enrique Tito, Edwin Contreras and Victor Tola (Facultad de Agronomia–UMSA, La Paz, Bolivia), Ivan Arnold (Fundación NATIVA, Tarija, Bolivia), Leopoldo Mamani (Secretaría de Desarrollo Agropecuario del Municipio de Independencia, Cochabamba, Bolivia) and Vito Salinas (SENASAG, Bolivia). This research was supported by DANIDA (Danish International Development Agency) under the Project ANDESCROP No. 104. Dan. 8-1203.

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